Programming for Information Systems

Orientation

Campus Food Rescue · 10 weeks · no programming background needed

Why learn programming when AI can write code?

- AI writes code fast. Some of it is good.
- Some of it **runs but is risky**: a hidden rule, a lost record, private data on a public screen.
- Someone has to **decide what the system should do** and **notice the risk**.

> This course: enough Python to **read, test and judge** code, and your own small system built with **your decisions**.

Three things every week

🧠 Think

One information-systems idea: people, data, rules, records, history, reports …

💻 Code

The Python you need for that idea, nothing more.

⚖️ Judge

One design principle, used on "Good or risky?" code.

The system: Campus Food Rescue

An information system = people + data + rules + process + output

Campus Food Rescue as an example

People

donors, coordinators, volunteers, partners

Data

offers: food, portions, hours left, status

Rules

approve first; 20+ portions needs a team

Technology

Python, files, a web page: the tools, not the system

Process

offer → approve → pick up → deliver

Output

a coordinator list, a public board

A day in the system

- A donor **offers** food: what, how many **portions**, how many **hours left**.
- The system **checks** the offer and refuses bad input.
- A **coordinator** (a person) approves and chooses a partner.
- Volunteers **pick up** and **deliver**. Every change goes into the **history**.
- The coordinator sees everything; the **public board** shows no private data.

You build it, one part per week

Ten weeks, one system

Each week: one Python topic, one design principle, one part of your system

Week 1
What is an IS?
Clear names

Week 2
Rules
Explicit rules

Week 3
Records
One record per thing

Week 4
Functions · Midterm
One job per function

Week 5
Input checks
Never trust input

Week 6
Lifecycle
Wrong states hard

Week 7
Saving & history
Never lose data

Week 8
Reports & screens
Right info, right person

Week 9
Assemble
Separate decisions

Week 10
Verify & present
Verify before trust

Your decisions make your system different

Week	Your decision
2	When does an offer need a **team**? What is **urgent**?
3	Which food **categories** do we accept?
5	Which offers are **refused** at the door?
6	Must a coordinator **approve** first?
7	What does the **history** keep?
8	What can the **public** see?

Ten principles

1	Clear names and units	6	Make wrong states hard to reach
2	Explicit rules	7	Never lose data
3	One record per real thing	8	Right information for the right person
4	One function, one job	9	Separate what changes from what stays
5	Never trust input	10	Verify before you trust

⚖️ Good or risky? A taste

Version A (from an AI assistant)

```
x = 30
y = 4
print(x / y)
```

Version B

```
portions = 30
volunteers = 4
each = portions // volunteers
print(f"{each} portions each")
```

How a week works

Session	What happens
1 · Learn	warm-up (research) · the idea · Python: explain, run, **predict**
2 · Practise and judge	short exercises · ⚖️ Good or risky?
3 · Build	write the part · ✅/❌ check · plug it in · 🧭 your decision

- ⭐ core for everyone · ⭐⭐ optional stretch
- **The minimum is always reachable:** run every cell and finish the ⭐ tasks.

Assessment (proposed)

Component	Weight
Participation (class + weekly reflection form)	10%
Six research tasks (5% each)	30%
Individual midterm (Week 4)	30%
Final project (Weeks 9–10)	30%

Submit by email: **submit.progmis@gmail.com** · deadlines and file names are announced in class.

Research tasks: half a page, real systems

	Topic	Share in
R1	A real information system	Week 2
R2	Rules and edge cases	Week 3
R3	Checking input	Week 5
R4	Data and history	Week 6
R5	Keeping data safe	Week 7
R6	Checking AI output	Week 8

Using AI tools

- ✅ Explore, ask questions, get explanations.
- ✅ Research tasks: allowed, if you **check a claim** and say how.
- ❌ The midterm: no AI tools.
- Always: you must be able to **explain** anything you submit.

> The course grades **understanding and judgement**, not the amount of code.

Getting started: Colab

1. Open the course website → your week → **Open in Colab**.
2. **File → Save a copy in Drive.** Every week, first.
3. Run ▶ **Run this cell first (setup)** → ✅ `Food Rescue system ready`.
4. Confused? **Runtime → Restart session and run all.**

Questions and help

- Ask in class, any time. Small questions help everyone.
- Good question = week + section + smallest code + expected vs actual.
- Private matters: **submit.progmis@gmail.com**
- Weekly reflection form: 3 minutes after each week.

Next: Week 1 · What is an information system?

Run the finished system, meet the people, write your first Python.